

PLT-1 Plant Sensor

Indoor plant multisensor — soil moisture, light, and climate for Home Assistant

The Apollo PLT-1 is an indoor plant multisensor for ESPHome and Home Assistant. It measures soil moisture with a durable capacitive probe, plus air temperature, humidity, light (LUX), and UV — giving you everything needed to automate watering, lighting, and climate for healthy plants. It runs locally on your network with no cloud or subscription required.

PLT-1 comes in a USB-C non-battery version and a larger battery version (rechargeable 18650 cell, not included). Onboard RGB and a piezo buzzer provide visual and audible plant-care alerts, and an optional waterproof probe adds soil-temperature monitoring. All firmware and CAD files are open source.

Key Specifications

Product	PLT-1 Plant Sensor
Manufacturer	Apollo Automation (Versailles, KY, USA)
Soil moisture	Capacitive sensor with conformal-coated ground contact
Air temp & humidity	AHT20-F
Light sensing	LTR-390UV — LUX + UV
Soil temp (optional)	DS18B20 waterproof probe, 20 cm (7.8 in)
Microcontroller	ESP32
Firmware	ESPHome — open source
Connectivity	Wi-Fi (2.4 GHz)
Power	USB-C (Non-Battery) or 18650 rechargeable (Battery model; cell not included)
Battery life	Est. ~6 months with default sleep/wake settings
Onboard feedback	RGB LED + piezo buzzer
In the box	Assembled sensor with case; Apollo sticker

Onboard & Optional Sensors

Sensor	Function	Availability
Capacitive soil probe	Soil moisture — conformal-coated for durability and accuracy	Standard
AHT20-F	Air temperature and humidity	Standard
LTR-390UV	Ambient light (LUX) and ultraviolet (UV)	Standard
DS18B20	Waterproof soil-temperature probe, 20 cm (7.8 in)	Optional

Configurations & Pricing

Configuration	Power	Price (USD)
Non-Battery (smaller form factor)	USB-C, continuous	\$27.99
Battery (battery not included)	18650 rechargeable, flat-top	\$29.99

Battery and charger are not included with the Battery model. Recommended cell sources: imrbatteries.com (USA), nkon.nl (EU).

Features & Use Cases

- **Automated watering:** trigger watering only when capacitive soil moisture drops below a set threshold.
- **Optimize sunlight:** use LUX and UV to automate shades or grow lights for the right light levels.
- **Climate management:** track air temperature and humidity to drive fans, heaters, or humidifiers.
- **Soil temperature:** add the optional waterproof probe to monitor soil warmth for sensitive plants.
- **Visual & audio alerts:** RGB LED for status (red = dry, green = optimal) and buzzer for urgent action.

Important Notes

- **Battery not included:** the Battery model needs a user-supplied rechargeable flat-top 18650 cell and a charger.
- **Battery life depends on sleep:** leaving "Prevent Sleep" enabled keeps Wi-Fi always on and sharply reduces battery life; with sleep enabled, expect roughly 6 months to a year depending on wake interval.
- **Form factor:** the Non-Battery version is significantly smaller than the Battery version.

Documentation & Resources

- **Product page:** apolloautomation.com/products/plt-1
- **Wiki / setup guide:** wiki.apolloautomation.com/products/plt1
- **Source & CAD files:** github.com/ApolloAutomation/PLT-1
- **Community:** [Apollo Automation Discord](https://discord.com/invite/apolloautomation)
- **Sensor datasheets:** Aosong AHT20-F, Liteon LTR-390UV, Maxim DS18B20.